

Subtracting Integers: Rule

Rule for Subtracting Integers

Our work with elevation spans demonstrated that **subtracting a number is equivalent to adding its opposite**. This rule is stated formally below.

Procedure 1 (Subtracting Integers). If a and b are integers, then $a - b = a + (-b)$.

Remark 1. Once the subtraction problem has been rewritten as addition, use the rules for adding integers to find the answer.

Example 1.

$$\begin{aligned} 6 - (-12) &= 6 + 12 \\ &= 18 \end{aligned}$$

Subtracting -12 is equivalent to adding positive 12.

Example 2.

$$\begin{aligned} 6 - 12 &= 6 + (-12) \\ &= -6 \end{aligned}$$

Subtracting 12 is equivalent to adding -12 .

Remark 2. When you subtract a positive number from a *larger* positive number, there is no reason to rewrite the subtraction problem as addition of the opposite. For example, you don't need to rewrite $7 - 2$ as $7 + (-2)$ to figure out that the answer is 5. So, **only rewrite subtraction as addition if you need to**.

Problem 1 Rewrite the subtraction problem as an equivalent addition problem. Then find the answer.

$$-6 - (-3)$$

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Multiple Choice:

(a) $-6 - (-3) = 6 + 3$

(b) $-6 - (-3) = -6 + (-3)$

(c) $-6 - (-3) = -6 + 3$ ✓

(d) $-6 - (-3) = 6 + (-3)$

Answer = -3

Problem 2 Rewrite the subtraction problem as an equivalent addition problem. Then find the answer.

$$-2 - 10$$

Multiple Choice:

(a) $-2 - 10 = 2 + (-10)$

(b) $-2 - 10 = 2 + 10$

(c) $-2 - 10 = -2 + (-10)$ ✓

(d) $-2 - 10 = -2 + 10$

Answer = -12

Problem 3 Rewrite the subtraction problem as an equivalent addition problem. Then find the answer.

$$5 - 8$$

Multiple Choice:

(a) $5 - 8 = -5 + 8$

(b) $5 - 8 = 5 + (-8)$ ✓

(c) $5 - 8 = -5 + (-8)$

(d) $5 - 8 = 5 + 8$

Answer = -3

Problem 4 Rewrite the subtraction problem as an equivalent addition problem. Then find the answer.

$$-16 - (-1)$$

Multiple Choice:

(a) $-16 - (-1) = -16 + 1$ ✓

(b) $-16 - (-1) = -16 + (-1)$

(c) $-16 - (-1) = 16 + 1$

(d) $-16 - (-1) = 16 + (-1)$

Answer = -15

Problem 5 Rewrite the subtraction problem as an equivalent addition problem. Then find the answer.

$$30 - (-10)$$

Multiple Choice:

(a) $30 - (-10) = 30 + (-10)$

(b) $30 - (-10) = 30 + 10$ ✓

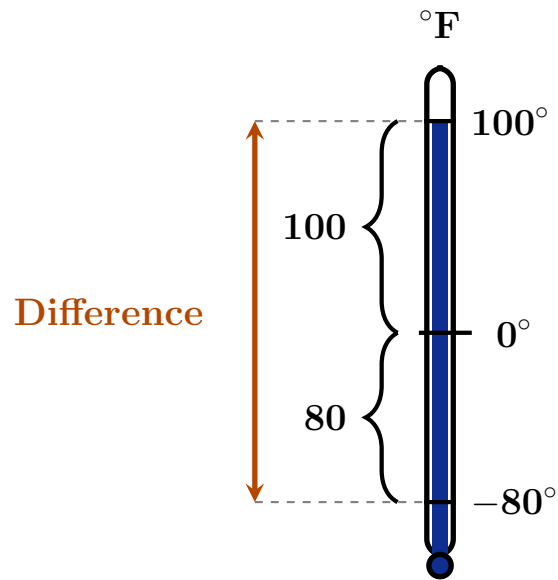
(c) $30 - (-10) = -30 + 10$

(d) $30 - (-10) = -30 + (-10)$

Answer = 40

Problem 6 How many degrees warmer is $100^{\circ} F$ than $-80^{\circ} F$?

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Alt text: A vertical thermometer diagram labeled in degrees Fahrenheit, illustrating the distance between negative 80 degrees and positive 100 degrees. The blue liquid inside the thermometer reaches up to the 100-degree mark. A solid black line marks the 0-degree point in the middle. On the left side, visual brackets measure the distances from negative 80 to 0 (labeled 80) and from 0 to 100 (labeled 100). Further to the left, a double-headed orange arrow spans the entire height from negative 80 to 100, labeled “Difference“ with dashed gray guide lines connecting it to the temperature marks.

Write the subtraction problem to find the answer.

Multiple Choice:

- (a) $100 - 80$
- (b) $100 - (-80)$ ✓
- (c) $-100 - 80$
- (d) $-100 - (-80)$

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Use the figure to rewrite the subtraction problem as an equivalent addition problem. Then find the answer.

Multiple Choice:

- (a) $100 + (-80)$
- (b) $-100 + 80$
- (c) $-100 + (-80)$
- (d) $100 + 80$ ✓

Answer = 180° F

Subtracting a Negative Number

You may have noticed that **subtracting a negative** number is equivalent to **adding a positive** number. This becomes intuitive when we think in terms of money. Imagine your checking account balance is \$900 and the bank realizes that they accidentally charged you a \$30 (-30) overdraft fee last month. Once the bank removes (subtracts) the overdraft fee, your net balance increases.

$$\begin{aligned} 900 - (-30) &= 900 + 30 \\ &= 930 \end{aligned}$$

Therefore, removing a debt is the same as gaining a profit.